
Art to Cheat the Common-Weale: Inventors, Projectors, and Patentees in English Satire, ca. 1630-70

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Art to Cheat the Common-Weale

Inventors, Projectors, and Patentees in English Satire,
ca.1630–70

JESSICA RATCLIFF

*“My brains I tosst, with many a strange vagary,
And (like a Spanniell) did both fetch and carry,
To you, such Projects, as I could invent,
Not thinking there would come a Parliament.*

...

*I had the Art to cheat the Common-weale,
And you had tricks and slights to passe the Seale.*

— John Taylor, 1641¹

On 3 February 1634 an extravagant masque, some say the most spectacular of the period, was staged by the Inns of Court for the entertainment of King Charles and Queen Henrietta.² “Invented and written” by James Shirley, designed by Inigo Jones, and titled *The Triumph of Peace*, the masque began with personifications of Confidence and Opinion leading a parade of lavishly costumed gentlemen on chariots and horses.³ But before this procession of lace and feather could advance very far, the mood was punctured by

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1. John Taylor, *The Complaint of M. Tenter-Hooke the projector*.

2. James Shirley, *The Triumph of Peace*. Unless otherwise noted, all seventeenth-century primary sources cited here were accessed at Early English Books Online (EEBO). On the masque’s spectacularity, see Herbert Arthur Evans, *English Masques*, 203–4; for the economic and political context of the masque as a whole, see Brent Whitted, “Street Politics.”

3. Shirley, *Triumph of Peace*, frontispiece.

the boisterous arrival of Phansie (Fancy—that is, imagination), who was then cajoled into creating an antimasque. Now came young law students hilariously dressed up as “beggars and cripples,” which according to one observer, “contrasted nicely with the noble music and gallant horses that went before, making the both of them more pleasing.”⁴ But Opinion complained of the ripe smell of the beggars, so Phansie sardonically offered to devise more “cleanely,” less “base and sordide” persons. “What think you,” Phansie asked, “of Projectors?”⁵ Thus began the “Antimasque of Projectors.”

The antimasque was composed of a series of projectors, each dressed to signify some kind of invention, dancing past the king and queen while making a show of asking the Crown for a patent. The first figure was “a Jockey,” riding a tiny horse fitted with a giant bridle. He sought a patent for a hollow bit through which cooling vapors were pumped to keep the horse from tiring. The next projector was “a Country fellow in a leather doublet and gray trunk Hose,” with a gear-work wheel upon his head and in his hand a sail, seeking a patent for a new way of threshing corn “without help of hands.” The third was “a grimme Philosophicall fac’d fellow” in a furred gown and with a glowing furnace on his head, who asked the Crown for a patent for a new kind of stove that endlessly produces heat. “A most Scholastick Project,” remarked Opinion as the projector danced past, “his feet follow the motion of his braine.” The next projector was dressed in “a case of black leather vast to the middle, and round on the top with glasse eyes, and bellows, under each arme”; this contraption was a diving suit with air-refreshing bellows and special goggles for searching out “[g]old or what ever jewels ha’beene lost, [i]n any River o’the World.” Another figure was dressed “like a Seaman,” and had invented a new ship design that “can melt huge Rockes to Jelly.”⁶ Just behind the last projector came a stream of street urchins, and the whole group was then chased away by a pair of huge mastiffs. Apparently it was all very funny and Queen Henrietta was especially amused.

According to Francis Bacon, “the introduction of noble discoveries [in-ventorum] seems to rank highest among human activities by a long way.” Or, as an early English translation of the same aphorism from Bacon’s *Novum Organon* would put it, “inventors of things” are due “heroick honor” as “Destroyers of Tyranny &c . . . *Inventions . . . are man’s Glory, they cause him to be a God to the rest of mankind.*”⁷ But early modern English writers did not always treat inventors with such reverence. Inventors and their inventions have long been a popular subject of satire (fig. 1). Literature from

4. Bulstrode Whitlocke, *Memorials of the English Affairs* . . . , 19.

5. Shirley, *Triumph of Peace*, 6.

6. Ibid., 2–3, 7–8; Whitlocke, *Memorials of the English Affairs* . . . , 19–20. The patent-begging dance is described in Whitlocke’s account, but is not the printed version of the masque.

7. Graham Rees, ed., *Oxford Francis Bacon*, vol. 11: *Novum Organon*, aphorism 129, p. 193; Francis Bacon, *The Novum Organon of Sir Francis Bacon*, 17.

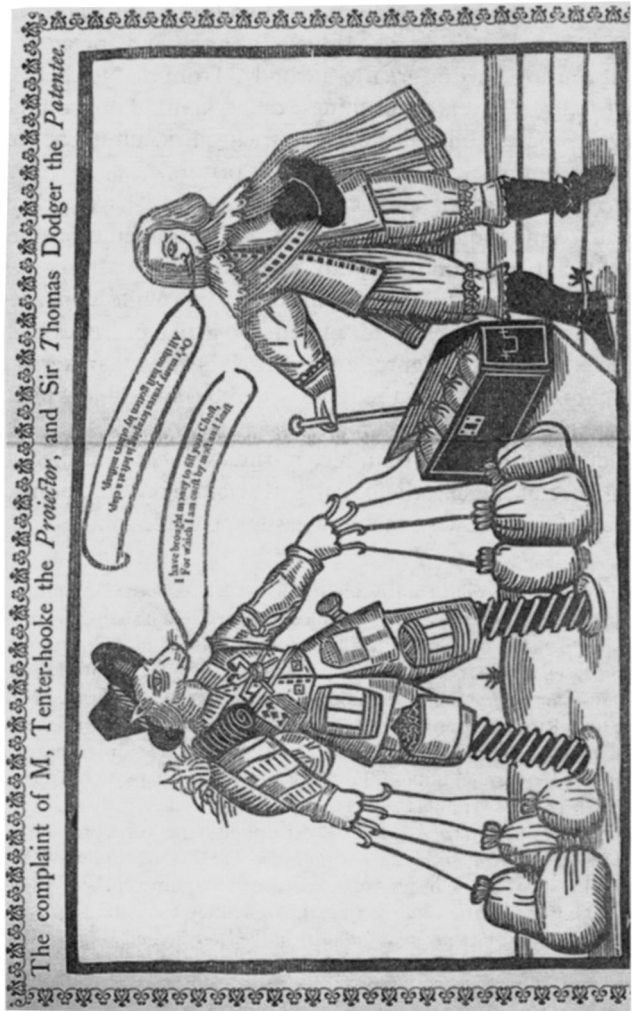


FIG. 1 A projector and a patentee, once partners in projects, now in financial ruin. Each blames the other, and the sitting of Parliament, for their sad state. On the left is the projector with the face of a swine, the ears of an ass, screw legs for his rising and falling status, and fish-hook fingers "to catch all trades." The projector's clothing is decorated with icons of notorious projects including tobacco, pipes, salt, cards, dice, wine, coal, and pins. On the right is the patentee, in gentleman's dress, guarding his chest of money. (Source: *Works of John Taylor* not included in the folio volume of 1630 [Spenser Society, C. E. Simms, 1870], courtesy Huntington Library, San Marino, CA.)

Ben Jonson to Jonathan Swift and beyond mocks and skewers the rhetoric of Bacon and his fellow promoters of invention. This essay focuses on satires of inventors as projectors in satirical plays, poems, and masques from the 1630s to the '70s. In these works, much of the critique is directed, in particular, at the rhetoric of invention as a route to social progress. To quote Bacon again: "The blessings of discovery can reach out to the whole human race . . . improvement of political conditions seldom proceeds without violence and disorder, whereas discoveries [inuenta] enrich and spread their blessings without causing hurt or grief to anybody." From the perspectives explored below, however, "inventors of things" called to mind not so much heroism or glory or progress, but a degraded get-rich-quick culture, oppression of the poor, corrupt royal power, and debtor's prison.⁸

Early modern conceptions of "projectors" and "projects" have been the subject of several recent studies.⁹ Most recently, Koji Yamamoto has produced an insightful cultural-economic study of the projector in the context of the financial revolution of the seventeenth century.¹⁰ While Yamamoto considers how the image of the projector played into actual practices of invention, this article explores the figurations themselves and the material of their construction in satire. During the last few decades the subject of science and literature has matured into an established interdisciplinary area. For the early modern period, important recent studies have established the interconnected nature of literary, rhetorical, and scientific forms, showing that "literature" and "science" were then not easily separable spheres.¹¹ But

8. Bacon quote from Rees, *Oxford Francis Bacon*, vol. 11: *Novum Organon*, aphorism 129, p. 193. Beyond the works discussed here, projector characters also appear in the satires of, for example, Ben Jonson (Sir Politick Wouldbe in *Volpone* [1607]; Meercraft, Lady Tailbush in *The Devil is an Ass* [1616]); Philip Messinger (The President of Projectors in *The Emperour of the East* [1632]); the unknown author of the lost play *Projector Lately Dead* (ca. 1630s); Richard Brome (Sir Andrew Mendicant in *The Court Beggar* [1632]; "projectors" in act 9 of *The Antipodes* [1638]); Margaret Cavendish (Monsieur Busie in *Wits Cabal* [1662]); Samuel Butler ("Projector" in *Characters* [written ca. 1667–69, published in 1759]); Thomas Wright (Lady Maurice in *The Female Virtuoso* [1693]); Jonathan Swift (Lord Peter in *A Tale of a Tub* [1704]); the philosophers and projectors in *Gulliver's Travels* [1726]); and William Hunt, *The Projectors* (1737). For studies of "anti-modern" figures in other areas of the history of science, see, for example, David Gentilcore, *Medical Charlatanism in Early Modern Italy*; Craig Ashley Hanson, *The English Virtuoso*; and Tara Nummedal, *Alchemy and Authority in the Holy Roman Empire*.

9. Koji Yamamoto, "Distrust, Innovation and Public Service"; J. M. Treadwell, "Jonathan Swift"; Joan Thirsk, *Economic Policy and Projects*; Christine MacLeod, *Inventing the Industrial Revolution*; and Maximillian E. Novak, ed., *The Age of Projects*. See also the session on "The Many Lives of the Projector: Inventors and Charlatans, Philosophers and Statesmen in Elizabethan and Stuart England" at the 2009 History of Science Society meeting in Phoenix.

10. In addition to Yamamoto's dissertation cited above, see also his "Piety, Profit and Public Service in the Financial Revolution."

11. For an excellent overview of recent work in this area, see Juliet Cummins and David Burchill, "Ways of Knowing." More generally, on literary studies of science and

while the rhetorical, cultural, and social dimensions of early modern science are now well-established, the subject of early modern technology remains much more historiographically isolated. Jonathan Sawday's recent "imaginative history" of the machine in early modern culture is a significant contribution to interdisciplinary histories of early modern technology.¹² So too, in a different way, is Lissa Roberts, Peter Dear, and Simon Schaffer's recent edited volume on "inquiry and invention" in the early modern period, which explores the ways in which "invention was as much mental as material."¹³ Debates about the nature of invention can be found across a wide range of early modern sources: from works of rhetoric to technical manuals to political tracts to poems. Historians have tended to isolate the subject of *technological* invention from the broader discourses of invention in which it is often embedded. There is therefore much more work to be done on the literary aspects of the history of technology and invention; for example, a study of the rhetorical aspects of the patent letter is long overdue.

This article examines literary aspects of the early modern discourse of invention and progress; it explores how the progressive rhetoric of promoters of invention are critiqued and lampooned in satire. The focus, in particular, is on the overlapping identities of inventor, innovator, and patentee as they are figured in the satirical character of the projector. Seventeenth-century satire was itself then considered one of the most inventive—if also base, lewd, and malicious—literary forms. It was a popular though loosely defined genre. Treatises on satire were written, but the genre had none of the formal elements of tragedy or comedy. Certainly there are signature elements of satire—for example, a derisive attitude toward the subject, a mocking or belittling tone, heavy application of sarcasm and irony—but there is little in the way of conventional plots or forms, or even the situation of the audience with respect to the work. Satire also openly plays out the biases of the satirist. "In satire," writes Michael Seidel, "the object of an action is identified primarily by the stance taken against it. The satirist depicts things as absurd, disreputable, or hypocritical because he deems them so."¹⁴ Critics like Seidel have characterized pre-Restoration satire as sporadic and spotty or ill-formed compared to that of the Restoration, when a strong current of court satire emerged. The early eighteenth century saw the rise of "great systems satires" that turned "a heavily localized and vitriolic

technology in the vernacular, see, for example, Philip Butterworth, *Magic on the Early English Stage*; John Shanahan, "Ben Jonson's *Alchemist* and Early Modern Laboratory Space"; Hillary M. Nunn, *Staging Anatomies*; Jonathan Sawday, *Engines of the Imagination*; Harry S. Turner, *The English Renaissance Stage*; and Hugh G. Dick, "The Telescope and the Comic Imagination."

12. Sawday, *Engines of the Imagination*, xv.

13. Lissa L. Roberts, "Introduction," 1–8.

14. Michael Seidel, "Satire, Lampoon, Libel, Slander," 33.

brand of satire based mainly on verbal tirade, pointed lampoons and libel into a more general attack on systems of related behaviors that encompass politics, aesthetics, religion, commerce and knowledge.”¹⁵

The satires explored in this article span the divide between pre- and post-Civil War culture. Their interconnections are a reminder of the many cultural continuities during this volatile period. They may also be regarded as early precursors to those later “system” satires of Daniel Defoe, Alexander Pope, and the Scriblerian Club. Although “projector” as a slanderous label certainly appears in court satires, the majority of projector satires are not focused on the court; rather, these satires present a derisive, cynical take on the patent system and, more generally, on the improvers’ rhetoric of progress.¹⁶ I employ the analytical category of “projector satires,” collecting a number of works that use the satirical character of a projector across a variety of forms. Not all of these works would be labeled primarily as works of satire, but each, at times, falls into satire with the description and action of the projector character. Shirley’s *Triumph of Peace*, for example, is not, on the whole, a satire, but the “Antimasque of Projectors” surely is.

This article explores the critical figurations of projectors in the context of the broad and shifting meaning of both “projector” and “inventor” in the early modern period. It was often claimed by promoters of inventions, projects, and patents (such as Francis Bacon), that their critics were motivated by a “siren song of reverence for antiquity.” Historians have tended to agree with Bacon on this point. But the three works analyzed below suggest a different way to understand early modern critics of improvers.¹⁷ Satires of inventors qua projectors are not expressions of a backwards-looking antediluvian or Aristotelian scholastic epistemology, but rather display a diverse response to the contemporary political and economic context for invention. The critique was generally not directed against technology or invention per se, but at the political and economic interests within which projects were then embroiled.

However, as the satires of projectors in the Inns of Court’s ostentatious royal masque should make clear, this critique should not be read as a picture of invention “from below”; there is no single target of these satires, and their meanings are unstable and, at times, contradictory. The term “projector” was slung in all directions. At times, the satires serve to displace distrust from one form of invention (the literary or rhetorical) to another (the material or technological). Yet there is a common ground to this set of critiques that is cultural, economic, and contemporary in nature.

15. Ibid., 50.

16. For example, the inventors Samuel Morland and Goodwin Wharton are both ridiculed as projectors in “An Answer to the Satire on the Court Ladies” (1680), which is reprinted in John H. Wilson, *Court Satires of the Restoration*, 43.

17. Rees, *Oxford Francis Bacon*, vol. 11: *Novum Organon*, aphorism 84, p. 133.

The Projector as Inventor

In the sixteenth century “invention” referred primarily to devices of rhetoric.¹⁸ From these literary origins the term had broadened by the seventeenth century to refer to new creations of all kinds. Invention also included discovering—as in uncovering or finding out—something that was locally unknown; for example, to bring a new foreign product into England was an act of invention, as was the creation of a new product or process by one’s own hands. New or locally unknown economic, political, or literary devices were equally described as inventions. In essence, the term “projector” meant creator, author, or inventor, in this broad, early modern sense of inventor. Throughout the seventeenth century the term was often used to describe anyone scheming for power, whether they were patentees, parliamentarians, Puritans, papists, or royalists.¹⁹ But the connotations were not all negative; God was sometimes the “Grand Projector,” and the term could simply be used, as it is in a religious pamphlet from 1626, to mean “devisor of a new plan.”²⁰

The projector or projectress as a popular character of satire emerged between 1600 and 1630, along with the genre of character study itself. One of the earliest book of characters, Joseph Hall’s *Book of Virtues and Vices* (1608), does not list the projector as a distinct character, but both “The Distrustfull” and “The Ambitious” are involved in “strange projects” and “projected plots.”²¹ One of the earliest stage satires to employ characters called projectors is found in Ben Jonson’s comedy *The Devil is an Ass* (1616), in which the question, “But what is a *Projector*?” is given the answer: “Why, one that projects wayes to enrich men, or to make ‘hem great, by suites, by marriages, by vndertakings.”²² Here, a projector is one who invents new ways to make money, whether by arranging a profitable marriage or investing in a profitable enterprise. Thus Jonson’s projectors are characters whose schemes are in one moment political and in the next technological—there is little distinction. As we will see, satires of projectors often depict technological invention as emblematic of the projector’s works.

Yet it is important to stress the broadness of the projector discourse, and to emphasize that *technological* invention was not then seen, as it is often seen today, as a distinct kind of invention with its own insulated evolutionary or uniquely progressive history.²³ Indeed, the terms “engine,” “device,” or

18. See Ullrich Langer, “Invention,” 136–45.

19. For a satire on the revolving political allegiances of the projector, see *The New Projector* (n.a.).

20. Thomas Scott, *The Projector*.

21. Quoted in Yamamoto, “Distrust, Innovation and Public Service,” 56.

22. Jonson, *The Devil is an Ass*, 1.7.10–14. For more on the projector character in this play, see Yamamoto, “Distrust, Innovation and Public Service,” 56–58.

23. For two prominent examples of evolutionary and exceptionalist approaches to

“instrument” could apply as equally to mental creations—for example, literary, rhetorical, or logical—as to objects or artifacts. And it was not unusual for individuals to work across this range of intellectual and technological forms. Consider, for example, the output of the self-styled projector Simon Sturtevant (ca.1570–1624?). Sturtevant’s first three works were literary: in 1597 he published a Latin grammar, and in 1602 two works advertised as “scholastic memory aides.”²⁴ In 1612 he turned to a rather different track: ironworking. Yet when he sought to gather investors’ confidence in his patent for a new kind of ironworking furnace, he did so by pointing to the success of his earlier “devices”: “The Inventioner [Sturtevant referring to himself] by his study, industrie, and practice, hath already brought to passe and published diuerse projects and new deuices, as well littoral as Mechannical, very Beneficiall to the common-wealth. His literare inventions . . . [include] *Dibre Adam a Scholasticall engin Aucomaton*.”²⁵

Still in the 1690s Defoe would refer to projectors as “authors” of inventions.²⁶ The breadth of the meaning and usages of invention, and the non-existence of a distinction between mechanical and other forms of invention, is central to appreciating the fuzzy categorical boundaries for the meaning and usage of projector.

In projector satires the association between invention and projecting is often direct. A projector is “infected with strange raptures and fancies, which he strives to put in practise, and calls them Projects . . . he is a maker of Newes as well as of New Inventions.”²⁷ A projector is “a man of good forecast . . . a creator of Court suites, an Inventor and devisor of new things, and a pretended reformer of the old.”²⁸ But these inventions are at once intellectual (for example, the idea of a butter stamp or of a monopoly on salt), political (the actual patent of monopoly), and material (for example, the purported new means of producing or processing salt described by the letter patent). At the same time, mechanical inventions in particular, which were often quite similar to those in the “Antimasque of Projectors,” regularly appear as a favorite accessory of the projector character. In one character study the projector “will turn all Wagons, Carts and Coaches in the manner of Windmills, to saile to their stations”; he “hath got a Patent to make wooden horses,” and has “an Art to convey himself . . . fifty fathom

the history of technology, see George Basalla, *The Evolution of Technology*; and Joel Mokyr, *The Lever of Riches*.

24. Simon Sturtevant, *Anglo-Latinus nomenclator* . . . ; *Dibre Adam, or Adams Hebrew dictionarie*; and *The etymologist of AEsops fables*. The only substantial study of Sturtevant is William H. Sherman, “Patents and Prisons.”

25. Sturtevant, *Metallica*. The entire text is reprinted in Bennett Woodcroft, ed., *Supplement to the Series of Letters Patent*, 4.

26. Daniel Defoe, *An Essay Upon Projects*, 11.

27. Thomas Heywood, *Hogs Character of a Projector*, 1.

28. Brugis, *Discovery of a Projector*, 2.

under water.”²⁹ Just as these satires sometimes draw on real monopolies, the frequent references to new carriages, devices for underwater exploration, water pumps, and new furnaces could have referred to any (or none) of several contemporary projects for such devices by inventors like Monsieur De Son, Samuel Morland, Cornelius Drebbel, and Kenelm Digby.³⁰

The Projector as Patentee

Bundled into the farcical “Antimasque of Projectors” in Shirley’s *Triumph of Peace* was a more serious, albeit gently composed political message. The year 1634 was a time of increasing conflict between the Crown and Parliament over patents and monopolies. Patents were granted not only for technological inventions, but also for all kinds of Crown privileges like titles, estates, and monopolies on the production and sale of commodities. Letters patent for invention had come into common use during the latter years of the reign of Elizabeth I.³¹ Obtaining a patent was expensive: the procedure—which basically continued unchanged until the nineteenth century—was a lengthy and circuitous ten-step process through the offices of Whitehall, where fees and gratuities were to be paid out to various officers.³² These officers did not investigate the invention itself or whether the patent infringed on any other patents or rights, but rather, if any issues were raised, they were more likely to be about how the patent would impact Crown revenue, especially the excise tax.³³

It may have been that, in theory, patents of invention were intended to benefit the public as well as the patentee, the Crown, and the economy in general. In fact, in practice the policies employed by monarchs from Eliza-

29. Heywood, *Hogs Character of a Projector*, 1.

30. De Son and Morland can be pointed to as inventors working in that period on different forms of spring-hung carriages. Submersible devices were reportedly devised by Morland (a diving bell for an individual), De Son (his “war machine” of Rotterdam), Papin (a submarine), and Cornelius Drebbel (a submarine with a means of chemically “refreshing” the air). On De Son, see Marika Keblusek, “Keeping it Secret”; on Drebbel, see L. E. Harris, “Cornelius Drebbel”; on Digby, see Bruce Janacek, “Catholic Natural Philosophy”; and on Morland, see J. R. Ratcliff, “Samuel Morland and His Calculating Machines c. 1666.”

31. For histories of the patent system in the early modern period, see MacLeod, *Inventing the Industrial Revolution*; Arthur Gomme, *Patents of Invention*; and William Hyde Price, *The English Patents of Monopoly*. The majority of work on patents in the history of technology focuses on the Industrial Revolution and after; for example, a special issue on patents in *Technology and Culture* (“Making Inventions Patent,” vol. 32, no. 4 [October 1991]) focused primarily on the late eighteenth and nineteenth centuries, with one important exception: Pamela O. Long’s “Invention, Authorship, ‘Intellectual Property,’ and the Origin of Patents.” See also Mario Biagioli, “From Print to Patents.”

32. MacLeod, *Inventing the Industrial Revolution*, 41.

33. *Ibid.*, 21.

both I to George III tended to treat patents primarily as a source of revenue for the state, basically as an extension of the Crown's right to excise and taxation. Under Elizabeth I there had initially been a sustained policy of granting letters patent for invention as a means of stimulating English manufacture and trade through the introduction of foreign products and methods. By the end of the sixteenth century, however, the power and presence of monopolies had expanded significantly; there were, for example, at least three different monopolies on salt, the initial ten-year term limit on monopolies had grown to thirty years, and the policy of granting patents only for new inventions had largely been abandoned.³⁴ Growing criticism led in 1601 to a parliamentary attack (and a defense by Bacon) on what was called an abuse of royal prerogative. In response Elizabeth revoked the most controversial patents. Yet under her successor, James I, the power of monopolies slowly built up to even greater heights than before, leading to another confrontation with Parliament in 1621. Like Elizabeth, James was forced to revoke a number of monopolies. This time the conflict with Parliament produced the Statute of Monopolies of 1624, which would set the basic form of the patent system until the nineteenth-century reforms.

As had his predecessors, Charles I found himself in increasingly desperate financial circumstances, which he met, in part, through the granting of patents and monopolies. The "Antimasque of Projectors" was but one of many calls for the Crown to recall and reform monopolies. Indeed, critical satires of patentees and monopolists—especially under the label "projectors"—were not at all new. To take just one example from Jonson in *The Devil is an Ass*, the "lady projectresse" Taile-Bush seeks a patent for a new kind of "paint," or make-up, that she claimed to have brought over from the Spanish court, but, in fact, had concocted out of toxic herbs.³⁵ The projectors in the "Antimasque of Projectors" would have been familiar characters to the audience, as would the underlying message. As Bulstrode Whitlocke—himself closely involved in the masque's production—would later recall: "Several other Projectors were in like manner personated in this Anti-masque; and it pleased the Spectators the more, because by it an Information was covertly given to the King, of the unfitness and ridiculousness of the Projects against the law."³⁶

Yet as much as the king and queen may have enjoyed the satire, they did not take this message much to heart. While Charles refused to recall patents or call Parliament to session throughout the 1630s, public grievances

34. Price, *The English Patents of Monopoly*, 7–8.

35. Ben Jonson, *The divell is an asse a comedie written in the year 1616, by his Majesties servants*, 3.4.27. The play also contains an extended satire on projects featuring Meercraft the projector who entraps a squire and his wife with the promise of the title of Duke and Duchess of Drowned-Land—a reference to the notorious Fens drainage projects.

36. Whitlocke, *Memorials of the English Affairs* . . . , 20.

against monopolies were left to fester. “The projectors swarmed again,” recalled the anonymous author of *The Anti-Projector* in 1646, “for they knew the King was irreconcilable to Parliaments.”³⁷ When it finally sat in 1640, topping the list of the Long Parliament’s grievances was the issue of monopolies. It was resolved that all “projectors and monopolists whatsoever . . . are disabled by order of this House to sit here in this House . . . he shall be dealt with as a stranger, that hath no power to sit here.”³⁸ This accounts for the common depiction in satire of projectors being afraid of Parliament. In a poem by John Taylor, the Water Poet from 1641, a projector “tosst” his “brains” to bring to a patentee “such projects as I could invent, not thinking there would come a Parliament.”³⁹ According to prolific playwright and poet Thomas Heywood (ca.1573–1641), a projector is “one whose arse makes buttons by the bushel at the sound of Parliament.”⁴⁰

Thus by the mid-1630s, as monopolies became a critical political issue, satires often conflated projectors with “patentees” and “monopolists” (two labels that themselves were then used interchangeably). This is the case for numerous satirical pamphlets and character studies published during the period of lax printing regulation in the lead-up to the Civil War, one of which will be explored in more detail below.⁴¹ By the early 1640s, as Christine MacLeod has pointed out, in the public imagination at least, patents and monopolies came to be implicated as a major cause of the Civil War.⁴² One pamphlet by Heywood begins with a poem sardonically lamenting the abuse of projectors at the sitting of the Long Parliament. “Machiaevell’s” (Machiavelli’s) ghost is risen to defend his “deare sons” the “Moderne Projectors”:

Why should your actions suffer censure, when
You were indeed the onely Men of men,
That did with cautious industrie supplie
Natures defects; and to Monopolie
Reduce all Trades, and Sciences within
The Kingdome, from the Beaver to the Pin.

In this text projectors are equated with the most reviled monopolies of James’s reign, including tobacco, rags, soap, salt, and butter. Like Catholics, they “raised profit out of sin” with their monopolies on wine, cards, and

37. *The Anti-Projector* (n.a.).

38. Quoted in Price, *The English Patents of Monopoly*, 46.

39. John Taylor, *The complaint of M. Tenter-hooke the proietor*.

40. Heywood, *Hogs Character of a Projector*, 6.

41. See, for example, Taylor, *The complaint of M. Tenter-hooke the proietor*; Brugis, *Discovery of a Projector*; and Heywood, *Machiaevell’s ghost and Hogs Character of a Projector*.

42. MacLeod, *Inventing the Industrial Revolution*, 16. Historians disagree about the precise role that patents played in the early modern economy, and there is even more debate about the role monopolies may have played in the build up to the Civil War. Price and Gomme stress the abusiveness, but MacLeod sees more balance.

dice.⁴³ This was the tail end of what Joan Thirsk described as the “scandalous age of projects.”⁴⁴

Restoration connotations of projecting retained much of the same meaning from the early Caroline era, while from the mid-1660s onward projectors can be found in a wider array of economic contexts. In John Wilson’s stage comedy *The Projectors* of 1665 the projector is still, ultimately, a patent-seeker pursuing courtier connections, but the play’s action concerns investment scams rather than monopolies. By the end of the seventeenth century the domain of the projector character seems to have definitively shifted from the Patent Office to Exchange Alley.⁴⁵ A cant and slang dictionary from the 1690s defines projectors broadly as “Busybodies in new Inventions and Discoveries, Virtuoso’s of Fortune, or Traders in unsuccessful if not impracticable Whimms, who are alwaies Digging where there is no more to be found.”⁴⁶ The question increasingly became one of distinguishing the “good” and the “bad” projector, as Defoe attempt to do in the 1690s in his *An Essay Upon Projects*. Swift’s “Academy” and “club” of projectors in *Gulliver’s Travels* (1726) were less monopolists than natural philosophers and their pursuits were absurd, but hardly were they the social menace of the mid-seventeenth-century projector.⁴⁷ Eventually, and especially with the Industrial Revolution of the late eighteenth century, inventors would come to be synonymous with the “good projectors” and, as MacLeod has documented, the discourse of the “heroic inventor” would begin to spread.⁴⁸ Its wake all but obliterated this earlier discourse of the inventor as “divelish projector.”

Satirizing “inventional progression” in 1641 and 1665

One of the more influential projector satires of the mid-seventeenth century was published in 1641 by a relatively unknown author, Thomas Brugis (ca.1620–51). He was a surgeon and the author of several medical texts.⁴⁹ It is noteworthy that Brugis, as a member of the small, self-consciously professional group of learned physicians (he called himself a “doctor of physicke” and seems to have been educated at the University of Paris), does not enter into direct confrontation with negative figurations of

43. Heywood, *Machiaavel’s ghost*.

44. The “scandalous phase” of 1580–1624, as opposed to the “constructive phase” of 1540–80; see Thirsk, *Economic Policy and Projects*, chap. 3. Most of the anti-projecting pamphlets from the 1630s through the ’70s are not literary satires, but attacks on specific individuals or projects. For more on these pamphlets, see Yamamoto, “Distrust, Innovation and Public Service,” chap. 1.

45. A similar pattern is described by Treadwell in “Jonathan Swift,” 446–49.

46. *A New Dictionary of the Canting Crew* (n.a.).

47. Swift, *Travels into several remote nations*, 29–30.

48. MacLeod, *Heroes of Invention*.

49. These include *The Marrow of Physicke* and *Vade Mecum*.

medical practitioners, such as those that were called “empirics,” “quacks,” or “mountebanks”; rather, he describes how the projector wreaks havoc on nearly every other profession but his own.⁵⁰ His *Discovery of a Projector* is a thirty-six-page pamphlet that was written, according to the author, many years before it was published, which may place it close to Shirley’s *Triumph of Peace* or perhaps even earlier. In five sections *Discovery of a Projector* tells the story of the birth, life, and death of a projector. Although the satire begins, as most of them do, with a preface claiming that his purpose was not to “defame any one person,” the pamphlet is tightly based upon Sturtevant’s *Metallica* of 1612.⁵¹ Sturtevant, the grammarian and schoolmaster turned ironmonger, obtained a patent of monopoly in 1611 for thirty-one years for iron-smelting furnaces that operated with coal or peat instead of wood. *Metallica* contains the patent letter, as well as a treatise on the art of “met-tallical” invention, comprising a curiously thorough taxonomy. This taxonomy includes not only all kinds of inventions related to metallurgy, but also phases of the invention process, aspects of financial support, types of models and experiments for testing, and characteristics by which an invention should be evaluated. *Metallica* is also an impassioned defense of invention as a divinely guided historical process of “inventional progression.”⁵²

Neither Sturtevant nor *Metallica* are ever directly named in *Discovery of a Projector*, but the satire is sharp and direct. Brugis’s first section—titled “What Is a Projector?”—draws nearly paragraph by paragraph upon the first few sections of *Metallica*. The satire amounts to an inversion of Sturtevant’s claims about the virtues of the projector. Sturtevant wants his work to “refute the erroneous folly of such shallow simple persons, who cannot abide any new inuention . . . they utterly distate both the projects and Inuentors, nay they say . . . it will neuer prooue good or come to passe.”⁵³ To this end he describes a taxonomic tree of invention, at the root of which is “heuretica”: the “art of inuention, teaching how to find the new, and to

50. Jenny Ward, “Brugis, Thomas”; Harold J. Cook, “Good Advice and Little Medicine.”

51. It is possible that *Metallica* was more broadly influential than has so far been recognized. Sections of an unpublished Swedish treatise from the 1660s, “On the Finding of Iron Ores, the Understanding of the Nature and kinds of Ore,” borrows from it. See Martyn C. Rady’s review of *Iron and Steel on the European Market in the 17th Century*. The reputation of *Metallica* may also have been expanded by references to it in two other published letters patent for the same sort of invention. See John Rovenson, *Metallica, but not that published by Simon Sturtevant*; and Dud Dudley, *Dud Dudley’s Metallum Martis* . . . (both are reprinted in Woodcroft, *Supplement to the Series of Letters Patent*). Patents for new smelting techniques, especially ones claiming not to require charcoal, were highly contentious from Sturtevant’s time through the Restoration. Sturtevant’s patent was revoked in 1612 and granted to Rovenson, before eventually being bought by the Dudley family. See “Dudley, Dud,” in *Oxford Dictionary of National Biography*.

52. Sturtevant, reprinted in Woodcroft, *Supplement to the Series of Letters Patent*, 24–25.

53. *Ibid.*, 18.

judge the old." Heureka divides into two aspects of invention: the "Scientiffick" involving "precepts generall to all liberall arts"; and the "Mechanick," which is "the art of the Inuentor."⁵⁴ In Brugis the projector is "a creator of Court suites, an Inventor and devisor of new things"; the projector gives fancy names to his works ("the better to gull the ignorant"), one of which is *heureka*. "A Mechanick Invention," continues Brugis, "is the art of the *Projector* which by effectuall Instruments and meanes bringeth forth some new visible worke *pretended* good and profitable to the Commonwealth."⁵⁵

Sturtevant gives the printing press as an example of a useful and beneficial mechanic art. With a patentee's due attention to the proprietary claims of a single inventor, he takes care to explain that although the press involves various crafts and artisans, it should be termed an "invention" in respect of the "author that devised them," namely Gutenberg. Sturtevant goes on to note that indeed all "arts, sciences, mysteries, trades, crafts, things and deuises" now existing in the commonwealth are also inventions. He stresses, however, the distinction between the "Inuentioneer"—the "first author" of these things—and the "secondary persons" who now employ or create them, who are not inventors themselves but "artificers" and "tradesmen" working in "occupations" and "professions."⁵⁶ While Sturtevant thus argues that projectors belong to an ancient line of inventors, Brugis wants to stress that projectors are unique to the times, a product of the corrupted age. Brugis therefore warns that projectors will "*claim* all Arts, Trades, Crafts, Sciences, Mysteries, Occupations, Professions Devises, and flights whatsoever were merely *Projects*, in respect of the Authors that devised them. . . . *Projectors* . . . pretend to mend all these and to make them both more beneficiall, and more serviceable."⁵⁷

This idea is further developed in the subsequent sections of *Discovery of a Projector*. Here, aspects of *Metallica* are wound into a loose plot charting the life of a fictional projector. A "citizen" finds himself struggling to make ends meet and recalls a cousin at court who owes him a small debt. Knowing his cousin has no actual money to pay him, the citizen still hopes that somehow this connection to the court can return him some benefit. So he pesters his cousin until eventually the courtier produces an offer to invest in an invention for a new kind of furnace, "an ingenious Art consisting of heate without the sight of fire and smoake," a new kiln or furnace designed such that it can never catch fire and emits no fumes or smoke and gives of a "sweete heat" that roasts, bakes, boils, malts and so on, to perfection.⁵⁸ These descriptions are taken directly from Sturtevant's patent letter in *Metallica*.

54. *Ibid.*, 17–18.

55. Brugis, *Discovery of a Projector*, 3 (emphasis in original).

56. Sturtevant, reprinted in Woodcroft, *Supplement to the Series of Letters Patent*, 18.

57. Brugis, *Discovery of a Projector*, 4.

58. *Ibid.*, 9.

The courtier convinces the citizen that the project is sound by employing the very same sorts of reasons Sturtevant gave his readers for trusting in his invention—namely, his testimony that he has witnessed successful models and experiments.

The citizen readily agrees to invest in the project, and, with that, the citizen is reborn as a projector. But that is only the beginning, for money is needed for various fees, experiments, and trials. Funds are collected from wealthy merchants, poor widows, and everyone in between. In the next section, titled “How Many Professions He Left to Become a Projector,” the courtier’s eloquent discourse on the financial promise and moral virtue of invention convinces members of a dozen trades—lawyers, clerks, merchants, fishmongers, “mettle” (metal) men, millers, brewers, bakers, and so on—that invention was now “the true way of thriving, protection and good forecast, whereupon they all agreed, that this was the onely way to become suddenly rich . . . thenceforth they took up the name *Projectors*, men of the best forecast, utterly disclaiming all their former Trades, Professions, Arts, Sciences, Mysteries, Crafts and Occupations, to all intents and purposes whatsoever.”⁵⁹ Thus with a physician’s gaze Brugis depicts projecting spreading like a contagious disease, as honest trades are abandoned in a rush to get rich quick.

Discovery of a Projector became something of a template for projector satire in the following decades. It was likely the basis for several character studies and pamphlets by Heywood.⁶⁰ Among other similarities, Heywood’s character study echoes Brugis in describing projecting as a deadly, communicable disease passed around by the projector’s deceptive words. More surprisingly, *Discovery of a Projector* would become a key source for one of the most substantial pieces of Restoration projector satire, Wilson’s *The Projectors*.⁶¹

At the start of the Civil War the Puritan majority had banned plays as a moral threat, and theatres would remain closed until the Restoration in 1660. They reopened with Charles II patronizing the foundation of two companies. Many of the plays were revivals of Renaissance works, including that of Jonson, and the projector character remained popular.⁶² Wilson

59. *Ibid.*, 19.

60. See Heywood, *Hogs Character of a Projector* and *Machiaevell’s ghost*.

61. On Wilson, see Kathleen Menzie Lesko, “Wilson, John,” and “A Rare Restoration Manuscript Prompt-Book”; M. C. Nahm, “John Wilson and His ‘Some Few Plays’”; and James Maidmant, *The Dramatic Works of John Wilson*. Among literary critics today Wilson does not attract much interest and of his plays, *The Projectors* has been studied the least. In one recent study of Restoration satire, *The Projectors* is described as a neglected “subversive comedy,” which “shows the seams of Stuart ideology, seams through which one can see elements of Restoration society that resist being stitched into one fabric.” See J. Douglas Canfield, *Tricksters & Estates*, 210, 254.

62. Samuel Pepys, for example, records going to see Jonson’s *Volpone*, in which Sir Politic Would-Be is introduced as a projector. After seeing the performance on the

(1626–96) was, in addition to being a playwright and poet, a lawyer of Lincoln's Inn, secretary to the Duke of York, and a semi-official propagandist for James II. There are conflicting accounts of the reception of *The Projectors*, but Wilson wrote in line with the popular Jonsonian style.⁶³ Clearly, his satirical approach, and even aspects of the plot of *The Projectors*, draws on works of Jonson like *The Alchemist* and *Volpone*.⁶⁴ Much of Jonson's work was focused on London itself and the new "publicke riot," as he once called it, of a swiftly urbanizing and increasingly capitalized society. As one critic reads Jonson, "the tools of the new man" were "disguise and rhetoric."⁶⁵ These are also the key attributes of the projector as presented by Wilson.

The main characters of *The Projectors* are Sir Gudgeon Credulous, "A Projecting Knight," Jocese, "A Courtier," his assistant Driver, and the following: Suckdry, "A Usurer," Squeeze, "An Exchange Broker," and Gotam, "A Citizen." These last three are "All in for Projects." In terms of the plot there are both romantic-economic and technological-economic projects in play: both Jocese and Sir Gudgeon seek to marry a wealthy widow, and Jocese also schemes to marry his son to the moneylender's daughter. The scene is contemporary London.

Jocese the courtier is in economic straits. Up to this point, as he says, he had always "gon the plain downright honest way." But times being as they are, he resolves to "follow the rest of the world" by turning to deceit, preying on the moneyed and gullible: "[I will] feed the humours of fools; and if they will sent up windmills in their heads, [I will] contribute my assistance to cutting out the sails."⁶⁶ Here is a typically Jonsonian moral set-up: "the vulnerability of all the characters could be seen as the result of social arrangements which prevent people from acting rightly."⁶⁷ As the play opens, then, Driver must take up the guise of a projector. "But can you cant?"—that is, can you speak like a projector—Jocese asks Driver? He answers:

Suppose I should tell [him—Gudgeon] I had studied the Emporeuticks, Lemnicks, Camnicks, and Plegnicks; could demonstrate the *minimum quod sit* of Homocrecious and Heterocrasious; and stripping Materia Prima to her smock, discover the most private recesses

evening of 14 January 1664/5, Pepys called it "a most excellent play: the best I think I ever saw."

63. Peggy Knapp, "Ben Jonson and the Publicke Riot." The centrality of a chaotic and changing London is also the focus of Heather C. Easterling's *Parsing the City*.

64. There are conflicting contemporary reports: "This play met with no great success" wrote Langbaine in 1699, but the editors of *Biographica Dramatica* (1812) state that the play "met with good success upon the stage"; both are quoted in Maidment, *The Dramatic Works of John Wilson*, 213–14.

65. Knapp, "Ben Jonson and the Publicke Riot," 177.

66. Wilson, *The Projectors*, 1.1.41.

67. Knapp, "Ben Jonson and the Publicke Riot," 173.

and occult qualities of Ignicadrillica, Metalorgonica, Euricatactica, and Hydropanta Pressoria? Do you believe, I say, [they] would be able to understand more of it than I do myself, which is just nothing?⁶⁸

To which Jocose replies: “Excellent! Then to subdivide them into undemonstrable yet seemingly probable projects. We shall make such sport!”

Here, then, is the essence of projecting: create an “undemonstrable yet seemingly probable” project and make it convincing by wrapping it in the language and apparent expertise of a science. Driver’s scientific canting—lemnicks, camnicks, plegnicks, and so on—is all drawn from Sturtevant’s taxonomy of invention in *Metallica*. Sturtevant’s carefully constructed system is here employed for pure comic effect and made to sound like nonsense. That Driver does not even understand what he is saying will not matter, because (it could be read two ways) either no one understands it, or else there is no real sense to it all anyway.

The first victim to be drawn into their schemes is the citizen Gotam, to whom Jocose is already in debt (in other words, here is the same setup as in Brugis’s *Discovery*). In a scene in which Driver runs into Gotam on the street and spills a bag full of “modells” (drawings) for his master’s inventions, Driver confides to Gotam that his master has developed “an ignike, hydrellick, and hydroterrlick invention, consisting of heat without fire or smoke!” (the same invention as featured in Brugis and originally described in Sturtevant’s patent). Driver then lets it slip that Jocose intends to obtain a patent: “My master, you know, has great friends, and therefore doubts not but by their assistance to procure a patent of privilege to engross a business solely to himself.” And, of course, because of the many fees to be paid to the patent clerks and in conducting experiments, it turns out that Jocose is taking on investors.

Upon hearing of the project, Gotam’s first thought is of the poor workers that the invention will displace: “Why, it will destroy all the woodmongers upon the river, and reduce them to their first dung-boats again!” And yet, after only a short pause, Gotam, now dreaming of securing a knighthood, is hooked. He has some savings to offer and even more: as warden of his parish church he offers to invest the parish stock. Thus the invention becomes a double blow to the poor: not only will it destroy their livelihood, but it will also divert parish funds into the project for that very destruction.⁶⁹ Note that this setup also illustrates how even if/when the project fails, it will still have done social damage by wasting capital on these schemes when it might otherwise have circulated more usefully within the community.

So Gotam is Jocose’s first victim. Eventually Driver succeeds in convincing each of his prey to invest sums adding up to £9,000 in the “ignicke, hydroterrlick engine,” but the projecting does not stop there. The romantic

68. Wilson, *The Projectors*, 1.1.51.

69. *Ibid.*, 1.1.101–10.

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projects continue and characters also approach Driver in secret, inquiring about other inventions in which they might become sole investors. As the plot continues, the introduction of each new project is met with a line about what livelihoods it will disturb, what modest industries it will replace. Driver ends up by gathering investments for not only the “*Metall-organicum Ignicadrillicum*,” but also, for example, for a tireless mechanical horse from Suckdry:

DRIVER: [showing him a “model” of the invention] Why ’tis a Wooden-horse, contriv’d with Screws and devices, that he shall out-travel a Dromedary, carry the burden of fifteen Camels, run you a thousand mile without drawing bit, and which is more then all this; not cost you two pence a year the keeping . . .

SUCKDRY: Ha, Ha, He—y’faith—y’faith! prythee on,—Is there no difficulty in the work?

DRIVER: The greatest, will be to set him a going; But I think I have sufficientlie provided for that:—I’ll tell you how I have order’d it: —Turn one Pin, he shall Trot, another Amble, a third Gallop; a fourth, Flie; And all this perform’d by *Germane* Clock-Work . . .—One thing more I could tell you—But—

SUCKDRY: Good Mr. *Driver*—out with it: —No Butts (among friends) I pray.

DRIVER: ’Tis but shoeing him with Cork, and he shall tread as firm, and strike as true a stroke on the water, as he does on land; and which is more, care for neither tyde nor weather, and run in the winds eye . . .⁷⁰

Driver also sells another invention to Gudgeon. This creation is a loom that can thread and weave single “atoms” so as to make cloth without wool (which will “alter the affairs of Christendom by breaking the Spanish trade of fine wool, and the Dutch new manufacture”).⁷¹ The atomic-loom project relies on a “glasse well ground” through which the atoms can be seen and caught up in a special needle. Upon hearing this Gudgeon becomes very excited, because he already has his own “glasse” through which he has discovered “a flea in the Bear’s tail, and a louse in . . . Medusa’s head.”⁷²

70. Ibid., 4.1.17–21.

71. Ibid., 2.1.205–15.

72. Ibid., 2.1.232. This would appear to be a timely and perhaps even last-minute reference to Robert Hooke’s *Micrographia*, which details observations of fleas, lice, and celestial bodies. *The Projectors* was granted an imprimatur in January 1664, *Micrographia* was published September 1664, and the play was first printed in 1665. Equally, however, the text could also be referencing other literature on microscopes and telescopes that refer to fleas and stars, such as Andrew Marvell’s poem “Upon Appleton House” (1650–52), stanza 53. Thanks to Kirsten Tranter for providing this reference.

Here, Wilson seems intent to draw some parallels between projecting and experimental philosophy—perhaps, though not necessarily, the recently founded Royal Society.

Some scholars have interpreted *The Projectors* as a satire of the early Royal Society.⁷³ However, while the play may satirize Gudgeon the dabbling experimental philosopher as a projector, that does not necessarily make a target of the Royal Society. Recall, for example, the frequent depictions—going as far back as the 1630s—of experiments, models, and trials as part of what a projector does. It seems that the experimental projector preceded the experimental philosopher, although in the public imagination there may have been little distinction between the two. We might rather read Gudgeon as just the sort of piffling, undisciplined experimentalist that the Royal Society was organized against. The more ridiculous the likes of Gudgeon, perhaps the more sensible seemed Fellows of the Royal Society.

As we might expect, one of the projector's most important tools is secrecy. Driver stalls for time and puts off demonstrations by making claims about the necessity of secrecy, and also by citing intractably occult qualities of his master's works.⁷⁴ Eventually, however, the investors' pressure to see some results forces Driver's hand and trials of the projects are scheduled. On the day that the inventions are to be revealed, however, Driver announces to each of the victims that the inventions have been confounded by unforeseen circumstances: the mechanical horse broke his leg, the atomic glass was tainted, and the furnace exploded at its first trial. The investors, of course, are furious, but resolution arrives when, at the end of the play, Driver's master Jocose appears with his new wife (the wealthy widow), chastises the men for having been taken in by the projects, has a good laugh at their greed and gullibility, and gives them all their money back. Everybody wins: Jocose, by way of gaining the widow for a wife; and the others, by learning the ridiculousness of their projecting ways. With a nod to *The Clouds*, Aristophanes' comedy of ideas that lampooned Socrates, the play closes as the characters address the audience in unison: "He who ploughs the clouds shall only reap the wind."

For all kinds of reasons, the dramatic and comedic potential of invention—especially technological invention—was a popular instrument for early modern playwrights, poets, and pamphleteers. Much of the drama lay in the political and economic entanglements in which inventions were snared; much of the comedy lay in the elements of fantasy and imagination that accompany the inventive process. But there was also the tragicomedy of rising and falling fortunes: inventors and innovators were depicted not only as a risk to the commonwealth, but also as a risk to themselves. Ac-

73. Lesko, "Wilson, John."

74. See, for example, Keblusek, "Keeping it Secret"; and Long, *Openness, Secrecy, Authorship*.

cording to Shirley, Brugis, Heywood, and Wilson, the patent office—at the intersection of politics, the economy, and the sciences, now the preferred domain of inventors of all kinds (for example, inventors of identities and ideas, as well as of technologies)—was a dangerous place to be. In both Brugis's and Heywood's character studies the projector falls ill while making his way through the endless series of offices in pursuit of his patent; as the fee paying and paperwork drag on his condition worsens. The death-blow comes when the projector, after passing out for a time, awakens to find his patent still unsealed. In Wilson's gentler and more upbeat satire, the would-be projectors were saved from this risk of personal ruin by the lessons learned from Jocose's pretended projecting. In reality, however, many a projector would have died a worse death in debtor's prison, where so many projectors ended their days. That, anyway, is where Sturtevant met his end, after ten years of incarceration following the failure of his project for a new kind of furnace.⁷⁵

The Invented Projector

Much as David Noble has stressed for a different time and place, in the satirical projector literature of the mid-seventeenth century the resistance to invention, innovation, or technological development explored here was not about technology or innovation per se, but about the politics of "progress."⁷⁶ These satires place into question rhetorics that presented innovation as progress, and progress as apolitical. *Discovery of a Projector* countered Sturtevant's rhetoric of the selfless patentee and the naturalness of inventional progression by discovering the projector to be a desperate courtier, or his poor relation, his inventions nothing more than fodder for investment schemes founded on the tangle of bureaucratic patronage that was the patent "system" at the time. In *The Projectors* a much more sprawling drama is built around the foibles of the projector character as it was laid out in *Discovery of a Projector*. The patent offices, and patents and monopolies in general, recede into the background as the drama surrounds a set of citizens becoming ensnared in technological scams, the first stage of a projected patent. Onstage the satire spreads a broader, in ways less complex critique: it ranges beyond the character of the projector—always gullible, gulling, and greedy—to the social consequences of projecting. The fashion for becoming rich "on a sodaine" is merely the starting point. The satire develops not upon character flaws, but upon the destructive ripples, through individual lives and society

75. On the high rate of incarceration of inventors in early-seventeenth-century London, see Sherman, "Patents and Prisons."

76. "For when technological development is seen as politics, as it should be, then the very notion of progress becomes ambiguous: what kind of progress? progress for whom? progress for what?" See David F. Noble, *Forces of Production*, xv, and *Progress Without People*.

at large, that those characters set in motion. Notably, much of this destruction occurs less in the patent system and more in the stock jobbing and investing associated with projects. In all these ways Wilson's reworking of Brugis indicates a somewhat different target of satire; it is significant that the duped investors are now not former craftsmen, but members of the financial trades. The scorn for the broker and the moneylender is nearly equal to the pretended projector who entraps them.

For all their differences, however, the overall thrust of these satires is the same: each presents a caustic take on the projector's rhetoric of innovative technological projects as progress for the public good. Historians have often interpreted such critical perspectives as being rooted in a historically oriented epistemology of Aristotelian scholasticism.⁷⁷ However, if anything, projectors are depicted as working against the social or economic order, not against orders of knowledge. It might therefore be suggested that critical satires of innovation should be interpreted in terms of a different aspect of scholasticism: Aristotelian ideas about wealth generation.⁷⁸ In the depiction of projectors as merely self-serving and only interested in accumulating wealth the literature could be read as marshalling the Aristotelian characterization of pure wealth-getting as unnatural and unethical against the projectors. Yet the self-styled moderns did not necessarily disagree with Aristotle's moral philosophy. It is easy to imagine a projector like Sturtevant fully agreeing with the Aristotelian distinctions on wealth-getting, and only disagreeing with the claim that his projects belonged to the unethical, self-serving form of moneymaking. That, of course, is the claim repeatedly made by projectors who would "help 1,000 of the poor" with their improvements. In other words, in much the same way that both sides of the enclosure debate argued that theirs was the solution to vagrancy, both improvers and their critics claimed to know what was best for the economy of the commonwealth and each employed ancient authority to their own rhetorical ends.⁷⁹

Thus although "ancients" and "moderns" are present, it is not an an-

77. For example, in his recent study of the debate surrounding the Fens drainage projects, Eric H. Ash argues for a reading along the ancient versus modern divide: "While some clung to an Aristotelian, teleological view of nature, others saw nature as a passive, disordered and manipulable entity that could (and should) be improved through carefully planned human intervention." See Ash, "Amending Nature," 121.

78. According to the first book of Aristotle's *Politics*, the natural and ethical generation of wealth must serve some external purpose or end (for example, benefiting the state or society) beyond that of simply wealth-getting or -generation in itself (for example, making a merchant wealthier). See Kimberly Latta, "Wandering Ghosts of Trade Whysies"; and Joao Luis Cesar Neves, "Aquinas and Aristotle's Distinction on Wealth."

79. Along the same lines, Canfield sees the ancients versus moderns dispute as being only superficially, or amphibiously, present in what he calls the "official discourse" of Restoration stage comedies: "under the guise of respect for Tradition, Restoration comedy . . . masks the aristocracy's alliance with the new science and trade." This is how, for example, he reads Thomas Shadwell's *The Virtuoso* (1676). See Canfield, *Tricksters & Estates*, 103, 109, 210.

cient-modern divide that motivated the satirical discourse described here; if anything, that rhetoric was itself lampooned as overly simplistic in these satires. As the opening lines of *The Projectors* would put it, "It is so hard to please, when things must be Mouldy with Age or Gilt with Novelty." Similarly, Heywood's projector "hugs himself with conceit of your ignorance, and his own wit. If you question him, his answer is, this Age is a cherisher of Arts and new inventions, the former dull and heavy."⁸⁰ In a light-handed fashion the satires also employ an inverted version of the ancient-modern divide. Certainly Bacon, Sturtevant, and other patentees and self-styled moderns often painted their critics as backwards-looking and outmoded. But in these satires inventors are themselves depicted as being too readily in step with the times; projectors are merely modish—they are slavish followers of cultural fashions.

It is equally difficult to situate the projector satires in terms of political motivations. Similar literatures have been described by Peter Burke as a "traditionalist" response to, or critique of, contemporary culture. The traditionalist response resists "in the name of the 'old order', changes which have been taking place. The emphasis may be on wicked individuals who break with tradition, but it may be on new customs (as we would say, new 'trends'). It is not a mindless conservatism but a bitter awareness that change is usually at ordinary people's expense, coupled with the need to legitimize riot or rebellion."⁸¹

This "bitter awareness that change is usually at ordinary people's expense" is clearly evident in the projector satires. Indeed, it could be argued that the projector's frequent claims of invention and innovation for the public good and poor relief are the most heavily satirized aspect of his character. Brugis, for example, says that the projector's "labor" amounts to "a Petition to his Majeste, with such mighty pretences of enriching the Kingdome . . . and [giving] employment for all the poore people of the Realme (which how well all these late *projects* have effected, I leave to judicious censure)."⁸² According to Heywood the projector's masterpiece is to propose a project that "tickles the ears of the king . . . caught in such terms, as might take the ignorant with applause, for all his pretences are preten[d]ed to the benefit of the King, the good of the Common-wealth, and the employment of 1000 of poor people; but [the g]ood man Never, thinks of any benefit for himselfe."⁸³ *The Projectors* also often reminds the audience of the dangers of projects, whether real or fraudulent, to the poor.

80. Heywood, *Hogs Character of a Projector*, 4–5.

81. Peter Burke, *Popular Culture in Early Modern Europe*, 175. For a recent analysis of Burke's work, see Matthew Dimmock and Andrew Hadfield, eds., *Literature and Popular Culture in Early Modern England*. On the emergence of the public as a politically and culturally important entity in seventeenth-century England, see Steve Pincus, "Coffee Politicians does Create."

82. Brugis, *Discovery of a Projector*, 2A.

83. Heywood, *Hogs Character of a Projector*, 2.

While aspects of the projector satires do fit Burke's traditionalist label, it is also clear that these works are not motivated by political radicalism, nor do they represent a picture of invention from below. If anything, the authors discussed above (namely, Brugis, Heywood, Taylor, and Wilson) tend to be politically conservative and are more or less connected to the court. Although the satirists attack the Crown's patent system and generally position themselves as defenders of the poor against predatory projectors and patentees, this is not, at root, a political critique.

Conclusion: The Projector as Poet

If any unifying motivation behind the projector satires can be identified, perhaps an element of it may lie in one factor they do have in common: that each walks a fine line by satirizing inventors via works of literary invention. A playwright with a new play and an engineer with a design for a new furnace would, in the early modern context, have been viewed as being more alike than they would be today. Both playwrights and inventors would seek patents of privilege to protect their works; both would be seen as "authors" of inventions. Satires of projectors in particular are especially harsh on schemers, pretenders, and imposters—characteristics that describe not only projectors, but also actors and authors. There is surely an element of self-awareness in the way these satires depict technological projects as merely a means of converting social capital to gold, when the authors were themselves employing their own inventions to those same ends.

The satirical projector character can, at times, be read as displacing or redirecting distrust from one set of inventors (the literary and rhetorical) toward another (the technological and material).⁸⁴ In Shirley's *Triumph of Peace* connections between literary and material invention are traceable on several levels. Masques were themselves sites of both literary and material invention. The scenic creations of Jones were especially well-known for their technical elements. Jonson famously dismissed Jones's masques as pandering to the "money-gett Mechanick Age."⁸⁵

The "Antimasque of Projectors" is presented as an invention within an invention, with "Phansie" seeming to devise the projectors on the spot. Even more, the "Antimasque of Projectors" is not as straightforwardly a critique of patents of invention as it might at first appear to be. According to Whitlocke the antimasque was the brainchild of Edward Noy, who was the then attorney-general.⁸⁶ Noy was closely involved in the granting of patents and

84. Thanks to Jenny Mann for useful comments on this point; see her *Outlaw Rhetoric*. See also Richard Halpern's use of displacement in his analysis of Thomas More's *Utopia* in *The Poetics of Primitive Accumulation*.

85. Jonson, "An Expostulation with Inigo Jones," line 52, quoted in Jean Agnew, *Worlds Apart*, 147.

86. See Whitted, "Street Politics."

monopolies, and in 1634 was widely despised for his support of the Westminster Soapboilers' monopoly. Thus, it seems, Noy presented projectors of submarines, newfangled furnaces, and perpetual motion as the truly ridiculous and unfit sorts of patentees, to be distinguished from such acceptable projects as that of the Soapboilers of Westminster. As it turned out, such an attempt to frame projects to suit his own purposes did not prevent Noy himself from being labeled a projector. Not too long after the *Triumph of Peace* he died a painful death due to kidney stones. According to several sources Noy's life and death soon became the subject of a stage comedy, now lost, produced in that same year, and he became the ridiculed projector. Titled *Projector Lately Dead*, the play depicted projectors covered in letters patent being stripped naked and ejected from the stage, and an autopsy of Noy that found a mass of hard white soap filling his body cavity.⁸⁷

Noy crossed a rather porous border from the patent offices into the literary world; plenty of others crossed that border in the reverse direction. Defoe was famously caught up in a civet cat-breeding project, and Swift proposed several projects for the moral and intellectual improvement of society.⁸⁸ There is no evidence that either Brugis or Wilson were ever similarly involved in projects. Still, both authors draw out similarities between poets and projectors, while, in a move similar to Noy's, depicting the projectors as the less trustworthy of the two. Brugis plays with this displacement when, in *Discovery of a Projector*, the first to be caught up in the citizen's project was a "cunning crafty Scrivener," a hack interrupted while "chopping up prayers." The scribe's enthusiasm should be no surprise, says Brugis, because members of this profession, with a great desire to become rich on a sodaine, are naturally inclined to projecting.⁸⁹ Wilson also invites his audience to consider the parallels between playwriting and projecting: in the prologue to *The Projectors* the audience is reminded that, even if the present play is a disappointment, there are other "projects" on offer. However, the prologue continues, be warned: unlike the dupes onstage, the playgoer's money is not going to be returned at the end of the show.

Bibliography

- Agnew, Jean. *Worlds Apart: The Market and the Theatre in Anglo-American Thought, 1550–1750*. New York: Cambridge University Press, 1986.
- Anon. *The Anti-Projector, or The History of the Fen Project*. S.l.: s.n., 1646. Early English Books Online (EEBO), available at <http://eebo.chadwyck.com/home> (hereafter EEBO).
- Ash, Eric H. "Amending Nature: Draining the English Fens." In *The Mindful*

87. See the University of Melbourne's lost plays database on *Projector Lately Dead*.

88. Treadwell, "Jonathan Swift."

89. Brugis, *Discovery of a Projector*, 13.

- Hand: Inquiry and Invention from the Late Renaissance to Early Industrialization*, edited by Lissa L. Roberts, Peter Dear, and Simon Schaffer, 117–43. Amsterdam: Edita, 2007.
- Bacon, Francis. *The Novum Organon of Sir Francis Bacon . . . translated out of the Latine by M.D.* London: Thomas Lee, 1676. EEBO.
- Basalla, George. *The Evolution of Technology*. Cambridge: Cambridge University Press, 1988.
- Biagioli, Mario. “From Print to Patents: Living on Instruments in Early Modern Europe.” *History of Science* 44 (2006): 139–86.
- Brome, Richard. *The Antipodes*. London: Francis Constable, 1638. EEBO.
- . *The Court Beggar . . . acted in 1632*. London: Richard Marriot and Tho. Dring, 1632. EEBO.
- Brugis, Thomas. *The Marrow of Physicke*. London: Richard Hearne, 1640. EEBO.
- . *Discovery of a Projector*. London: Lawrence Chapman and William Cooke, 1641. EEBO.
- . *Vade Mecum: or, a companion for a chirurgion, fitted for sea, or land*. London: Thomas Williams, 1651. EEBO.
- Burke, Peter. *Popular Culture in Early Modern Europe*. London: Maurice Temple Smith Ltd., 1978.
- Butterworth, Philip. *Magic on the Early English Stage*. Cambridge: Cambridge University Press, 2005.
- Canfield, J. Douglas. *Tricksters & Estates: On the Ideology of Restoration Comedy*. Lexington: University Press of Kentucky, 1997.
- Cavendish, Margaret. *Plays Written by . . . Lady Marchioness of Newcastle*. London: John Martyn, James Allestry, Tho. Dicas, 1662. EEBO.
- Cook, Harold J. “Good Advice and Little Medicine: The Professional Authority of Early Modern English Physicians.” *Journal of British Studies* 33 (1994): 1–31.
- Cummins, Juliet, and David Burchill. “Ways of Knowing: Conversations between Science, Literature and Rhetoric.” In *Science, Literature and Rhetoric in Early Modern England*, edited by Juliet Cummins and David Burchill, 1–12. Farnham, UK: Ashgate Publishing, 2007.
- Defoe, Daniel. *An Essay Upon Projects*. London: Thomas Cockerell, 1697. EEBO.
- Dick, Hugh G. “The Telescope and the Comic Imagination.” *Modern Language Notes* 58 (1943): 544–48.
- Dimmock, Matthew, and Andrew Hadfield, eds. *Literature and Popular Culture in Early Modern England*. Farnham, UK: Ashgate Publishing, 2009.
- Dudley, Dud. *Dud Dudley’s Metallum Martis . . . S.l.: s.n., 1665*. EEBO.
- Easterling, Heather C. *Parsing the City: Jonson, Middleton, Dekker, and City Comedy’s London as Language*. New York: Routledge, 2007.
- Gentilcore, David. *Medical Charlatanism in Early Modern Italy*. Oxford: Oxford University Press, 2006.

- Gomme, Arthur. *Patents of Invention: Origin and Growth of the Patent System in Britain*. London: Longmans Green and Co., 1946.
- Greenberg, Robert, and William Piper, eds. *The Writings of Jonathan Swift*. New York: W. W. Norton, 1976.
- Halpern, Richard. *The Poetics of Primitive Accumulation: English Renaissance Culture and the Genealogy of Capital*. Ithaca, NY: Cornell University Press, 1991.
- Hanson, Craig Ashley. *The English Virtuoso: Art, Medicine, and Antiquarianism in the Age of Empiricism*. Chicago: University of Chicago Press, 2009.
- Harris, L. E. "Cornelius Drebbel: A Neglected Genius of Seventeenth-Century Technology." *Transactions of the Newcomen Society* 31 (1957–59): 195–204.
- Heywood, Thomas. *Machiauevel's ghost, as he lately appeared to his deare sons the moderne projectors divulged for the pretended good of the kingdomes England, Scotland, and Ireland*. London: Francis Constable, 1641. EEBO.
- . *Hogs Character of a Projector, wherein is disciphered the manner and shape of that vermine*. London: G. Tomlinson, 1642. EEBO.
- Hunt, William. *The Projectors*. London: T. Cooper, 1737.
- Janacek, Bruce. "Catholic Natural Philosophy: Alchemy and the Revivification of Sir Kenelm Digby." In *Rethinking the Scientific Revolution*, edited by Margaret J. Osler, 89–110. Cambridge: Cambridge University Press, 2000.
- Jonson, Ben. *The Devil is an Ass a comedie acted in the year 1617*. London: s.n., 1641. EEBO.
- . *Volpone, or The Foxe*. London: George Elder, 1607. EEBO.
- . *The Complete Poetry of Ben Jonson*, edited by William B. Hunter Jr. New York: New York University Press, 1963.
- Keblusek, Marika. "Keeping it Secret: The Identity and Status of an Early-Modern Inventor." *History of Science* 43 (2005): 37–56.
- King, P. W. "Dudley, Dud (1600?–1684)." In *Oxford Dictionary of National Biography*, Oxford University Press, 2004; online edition, January 2008 (<http://www.oxforddnb.com/view/article/8146>).
- Knapp, Peggy. "Ben Jonson and the Publicke Riot: Ben Jonson's Comedies." In *Staging the Renaissance*, edited by David Scott Kastan and Peter Stallybrass, 65–79. New York: Routledge, 1991.
- Langer, Ullrich. "Invention." In *The Cambridge History of Literary Criticism*, vol. 3, edited by Glyn P. Norton, 136–45. Cambridge: Cambridge University Press, 1999.
- Latta, Kimberly. "'Wandering Ghosts of Trade Whymies': Projects, Gender, Commerce, and Imagination in the Mind of Daniel Defoe." In *The Age of Projects*, edited by Maximillian Novak, 141–65. Toronto: University of Toronto Press, 2008.
- Lesko, Kathleen Menzie. "A Rare Restoration Manuscript Prompt-Book:

- John Wilson's *Belphegor*, Corrected by the Author." *Studies in Bibliography* 32 (1979): 215–19.
- . "Wilson, John (bap. 1626, d. 1695?)." In *Oxford Dictionary of National Biography*, Oxford University Press, 2004; online edition, January 2008 (<http://www.oxforddnb.com/view/article/29663>).
- Long, Pamela O. "Invention, Authorship, 'Intellectual Property,' and the Origin of Patents: Notes toward a Conceptual History." *Technology and Culture* special issue on "Patents and Inventions," vol. 32, no. 4 (October 1991): 846–84.
- . *Openness, Secrecy, Authorship: Technical Arts and the Culture of Knowledge from Antiquity to the Renaissance*. Baltimore: Johns Hopkins University Press, 2001.
- MacLeod, Christine. *Inventing the Industrial Revolution: The English Patent System, 1660–1800*. Cambridge: Cambridge University Press, 1988.
- . *Heroes of Invention: Technology, Liberalism and British Identity, 1750–1914*. Cambridge: Cambridge University Press, 2007.
- Maidmant, James. *The Dramatic Works of John Wilson*. Edinburgh: W. Patterson, 1873.
- Mann, Jenny. *Outlaw Rhetoric: Figuring Vernacular Eloquence in Shakespeare's England*. Ithaca, NY: Cornell University Press, 2012.
- "The Many Lives of the Projector: Inventors and Charlatans, Philosophers and Statesmen in Elizabethan and Stuart England." Session at the 2009 History of Science Society Annual Conference, Phoenix (http://www.hssonline.org/Meeting/2009HSSMeeting/2009_Meeting_Program.pdf).
- Massinger, Philip. *The Emperor of the East*. London: John Waterson, 1632. EEBO.
- Mokyr, Joel. *The Lever of Riches: Technological Creativity and Economic Progress*. Oxford: Oxford University Press, 1992.
- Nahm, M. C. "John Wilson and His 'Some Few Plays.'" *Review of English Studies* 14 (1938): 143–54.
- Neves, Joao Luis Cesar. "Aquinas and Aristotle's Distinction on Wealth." *History of Political Economy* 32 (2000): 649–57.
- The Nevv Projector; or The Privileged Cheat*. London: William Gilbertson, 1662(?). EEBO.
- A New Dictionary of the Canting Crew in its several Tribes of Gypsies, Beggers, Thieves, Cheats &c . . .* London: s.n., 1699. EEBO.
- Noble, David F. *Forces of Production: A Social History of Industrial Automation*. New York: Alfred A. Knopf, 1984.
- . *Progress Without People: In Defense of Luddism*. Chicago: Charles H. Kerr Publishing, 1993.
- Novak, Maximillian E., ed. *The Age of Projects*. Toronto: University of Toronto Press, 2008.
- Nummedal, Tara. *Alchemy and Authority in the Holy Roman Empire*. Chicago: University of Chicago Press, 2007.

- Nunn, Hillary M. *Staging Anatomies: Dissection and Spectacle in Early Stuart Tragedy*. Farnham, UK: Ashgate Publishing, 2005.
- Pincus, Steve. "'Coffee Politicians does Create': Coffeehouses and Restoration Political Culture." *Journal of Modern History* 67 (1995): 807–34.
- Price, William Hyde. *The English Patents of Monopoly*. Cambridge, MA: Harvard University Press, 1913.
- Rady, Martyn C. "Review: Gunner Pipping, ed., *Iron and Steel on the European Market in the 17th Century: A Contemporary Swedish Account of Production Forms and Marketing*." *European History Quarterly* 16 (1986): 260–61.
- Ratcliff, J. R. "Samuel Morland and His Calculating Machines c. 1666: The Early Career of a Courtier-Inventor in Restoration London." *British Society for the History of Science* 40 (June 2007): 159–79.
- Rees, Graham, ed. *The Oxford Francis Bacon*, vol. 11. Oxford: Clarendon Press, 1996.
- Roberts, Lissa L. "Introduction." In *The Mindful Hand: Inquiry and Invention from the Late Renaissance to Early Industrialization*, edited by Lissa L. Roberts, Peter Dear, and Simon Schaffer, 1–8. Amsterdam: Edita, 2007.
- Rovenson, John. *Metallica, but not that published by Simon Sturtevant*. S.l.: s.n., 1612. EEBO.
- Sawday, Jonathan. *Engines of the Imagination: Renaissance Culture and the Rise of the Machine*. Abingdon, UK: Routledge, 2007.
- Scott, Thomas, B.D. *The Projector*. London: s.n., 1623. EEBO.
- Seidel, Michael. "Satire, Lampoon, Libel, Slander." In *The Cambridge Companion to English Literature: 1650–1740*, edited by Steven N. Zwicker, 33–57. Cambridge: Cambridge University Press, 1998.
- Shanahan, John. "Ben Jonson's *Alchemist* and Early Modern Laboratory Space." *Journal for Early Modern Cultural Studies* 8 (2008): 35–66.
- Sherman, William H. "Patents and Prisons: Simon Sturtevant and the Death of the Renaissance Inventor." *Huntington Library Quarterly* 72 (2009): 239–56.
- Shirley, James. *The Triumph of Peace: A Masque, Presented by the Four Honourable Houses, Or Innes of Court . . . February the Third, 1633*. London: John Norton, 1634. EEBO.
- Smith, J. *Diary and Correspondence of Samuel Pepys . . .*, vol. 2. Philadelphia: McKay, 1887.
- Sturtevant, Simon. *Anglo-Latinus nomenclator . . .* London: T. Gellibrand, 1597. EEBO.
- . *Dibre Adam, or Adams Hebrew dictionarie: a rare and new invention . . .*, 2nd ed. London: F. Kingston, 1602. EEBO.
- . *The etymologist of AEsops fables*. London: Robert Dexter, 1602. EEBO.

- . *Metallica*. London: George Elder, 1612. EEBO.
- Swift, Jonathan. *Travels into several remote nations . . . by Lemuel Gulliver*. Dublin: Risk, Ewing and Smith, 1727. EEBO.
- Taylor, John. *The complaint of M. Tenter-hooke the projector, and Sir Thomas Dodger the patentee*. London: Francis Coles, 1641. EEBO.
- Thirsk, Joan. *Economic Policy and Projects: Development of a Consumer Society in Early Modern England*. Oxford: Clarendon Press, 1978.
- Treadwell, J. M. "Jonathan Swift: The Satirist as Projector." *Texas Studies in Literature and Language* 17 (1975): 439–60.
- Turner, Harry S. *The English Renaissance Stage: Geometry, Poetics, and the Practical Spatial Arts, 1580–1630*. Oxford: Oxford University Press, 2006.
- University of Melbourne, Faculty of Arts. Lost plays database (2009), on *Projector Lately Dead*, available at http://www.lostplays.org/index.php/Projector_Lately_Dead.
- Ward, Jenny. "Brugis, Thomas (b. in or before 1620, d. in or after 1651)." In *Oxford Dictionary of National Biography*, Oxford University Press, 2004; online edition, January 2008 (<http://www.oxforddnb.com/view/article/3769>).
- Whitlocke, Bulstrode. *Memorials of the English Affairs . . .* London: Nathaniel Ponder, 1682. EEBO.
- Whitted, Brent. "Street Politics: Charles I and the Inns of Court's Triumph of Peace." *The Seventeenth Century* 24 (2009): 1–25.
- Wilson, John. *The Projectors. A Comedy*. London: John Playfare and William Crook, 1665. EEBO.
- Wilson, John H. *Court Satires of the Restoration*. Columbus: Ohio State University Press, 1976.
- Woodcroft, Bennett, ed. *Supplement to the Series of Letters Patent and Specifications of Letters Patent for Inventions*, vol. 1. London: Great Seal Patent Office, 1858.
- Wright, Thomas. *The Female Virtuoso*. London: R. Vincent, 1693. EEBO.
- Yamamoto, Koji. "Distrust, Innovation and Public Service: 'Projecting' in Seventeenth- and Early Eighteenth-Century England" (Ph.D. diss., University of York, 2009).
- . "Piety, Profit and Public Service in the Financial Revolution." *English Historical Review* 126 (2011): 806–34.